Here, the number 18 is a multiple of both 6 and 9 and so 18 is the first common denominator of both the fractions.

$\frac{5}{6} = \frac{5 \times 3}{6 \times 3} = \frac{15}{18}$ | $\frac{7}{9} = \frac{7 \times 2}{9 \times 2} = \frac{14}{18}$

Thus, $\frac{15}{18}$ and $\frac{14}{18}$ are like fractions, respectively equivalent to $\frac{5}{6}$ and $\frac{7}{9}$.

Example 6.7

Convert $\frac{4}{8}$ and $\frac{5}{16}$ into like fractions.

Solution

As 16 is twice 8, it is easy to make 16 as the common denominator.

$\frac{4}{8} = \frac{4 \times 2}{8 \times 2} = \frac{8}{16}$

Thus $\frac{8}{16}$ and $\frac{5}{16}$ are the required like fractions.

Example 6.8

Find a common denominator for $\frac{2}{5}$ and $\frac{3}{7}$.

Solution

The number 35 is a common multiple of both 7 and 5.
So, take 35 as the common denominator.

$\frac{2}{5} = \frac{2 \times 7}{5 \times 7} = \frac{14}{35}$ | $\frac{3}{7} = \frac{3 \times 5}{7 \times 5} = \frac{15}{35}$

Therefore $\frac{14}{35}$ and $\frac{15}{35}$ are the required like fractions.

Exercise 6.3

1. Convert the following into like fractions.

(i) $\frac{1}{4}, \frac{3}{8}$ (ii) $\frac{2}{5}, \frac{1}{7}$ (iii) $\frac{2}{5}, \frac{3}{10}$ (iv) $\frac{2}{7}, \frac{1}{6}$

(v) $\frac{1}{3}, \frac{3}{4}$ (vi) $\frac{5}{6}, \frac{4}{5}$ (vii) $\frac{1}{8}, \frac{3}{7}$ (viii) $\frac{1}{6}, \frac{4}{9}$

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